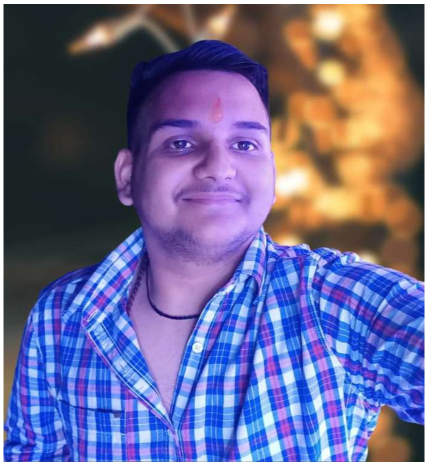
PROJECT AND TEAM INFORMATION

Project Title

SamaySudarshan /Timetable Wizard

Student / Team Information

Team Name: Team #	404 Found
Team member 1 (Team Lead) (Last Name, name: student ID: email, picture):	Gupta , Ashish – 23021345 ashishguptaa200425@gmail.com 

Team member 2 (Last Name, name: student ID: email, picture):	Lohani , Sameer – 23021048 sameerlohani200510@gmail.com 
Team member 3 (Last Name, name: student ID: email, picture):	Kimothi , Aditya – 23021776 adityakimothi31@gmail.com 
Team member 4 (Last Name, name: student ID: email, picture):	Dobhal , Varun – 23291039 varundobhal33@gmail.com 

PROPOSAL DESCRIPTION

Motivation

- Every new semester at our college, creating a timetable is a big headache for everyone. The current manual process almost always leads to classes clashing, which is a huge problem for both students and teachers. When classes overlap, students are forced to miss important lectures, and the whole learning process gets disrupted.

- But the problem isn't just about clashes. The schedules often feel unfair—some students finish their day early, while others are stuck on campus until late, with long, wasted gaps in between. This uneven workload is frustrating and makes it hard to manage the day. It also means that our classrooms and labs aren't being used as efficiently as they could be.

- The manual system is simply not keeping up, and it's causing a lot of frustration for everyone on campus. To solve this, we need a smarter, automated, and flexible solution. This kind of system won't just prevent clashes but it will also create a fairer schedule, use our college resources properly, and lead to a much better academic experience for all.

State of the Art / Current solution

- Right now, the process of creating a college timetable is a manual and difficult task. It's usually handled by a small committee or a few staff members who rely on spreadsheets or very basic software. They have to manually enter all the data—like classroom availability, faculty schedules, and course requirements—and then try to arrange everything by hand.

- This approach is extremely time-consuming and prone to errors. Since these tools can't automatically account for complex issues like a faculty member's preferred day off or a specific classroom's capacity, the final timetable often has conflicts. It's almost impossible to get a perfect schedule this way. The result is an inefficient system that leads to clashing classes, underused resources, and an uneven distribution of workload among teachers. Ultimately, this traditional, manual method is the main reason for all the scheduling problems we face every semester.

Project Goals and Milestones

Our main goal is to develop a complete, web-based platform called **SamaySudarshan/Timetable Wizard** that automatically generates optimized timetables for our college. We want this system to not only fix the current issues of class clashes and inefficient resource use, but also to create a fairer and more balanced schedule for students and faculty.

Our project will be built in three main phases, with specific milestones for each.

- **Phase 1: Foundation (Month 1)**
  The first month is all about building the project's solid foundation. Our main goal is to create a basic system that can take in data and produce a simple timetable, even without all the advanced features.
  - **Milestone 1:** By the end of this month, we'll have the project's core setup ready. This includes building the backend, designing a basic database to store information about faculties, students, and classrooms, and setting up the login system. We'll also develop a simple, rule-based algorithm that can generate a **clash-free timetable** based on just the hard constraints, like room availability. This is our first major step.

- **Phase 2: Intelligence & Optimization (Month 2)**
  This is the most crucial month where we'll add the brain to our project. We'll shift our focus from a basic system to a smart one.
  - **Milestone 2:** Our goal is to replace the simple algorithm from Phase 1 with a powerful **AI-powered optimization algorithm** (like a Genetic Algorithm). We will train this algorithm to handle soft constraints, such as a teacher's preferred day off, and to find the most balanced schedule. By the end of this month, the system will be able to generate and display multiple optimized timetable options for the admin to choose from.

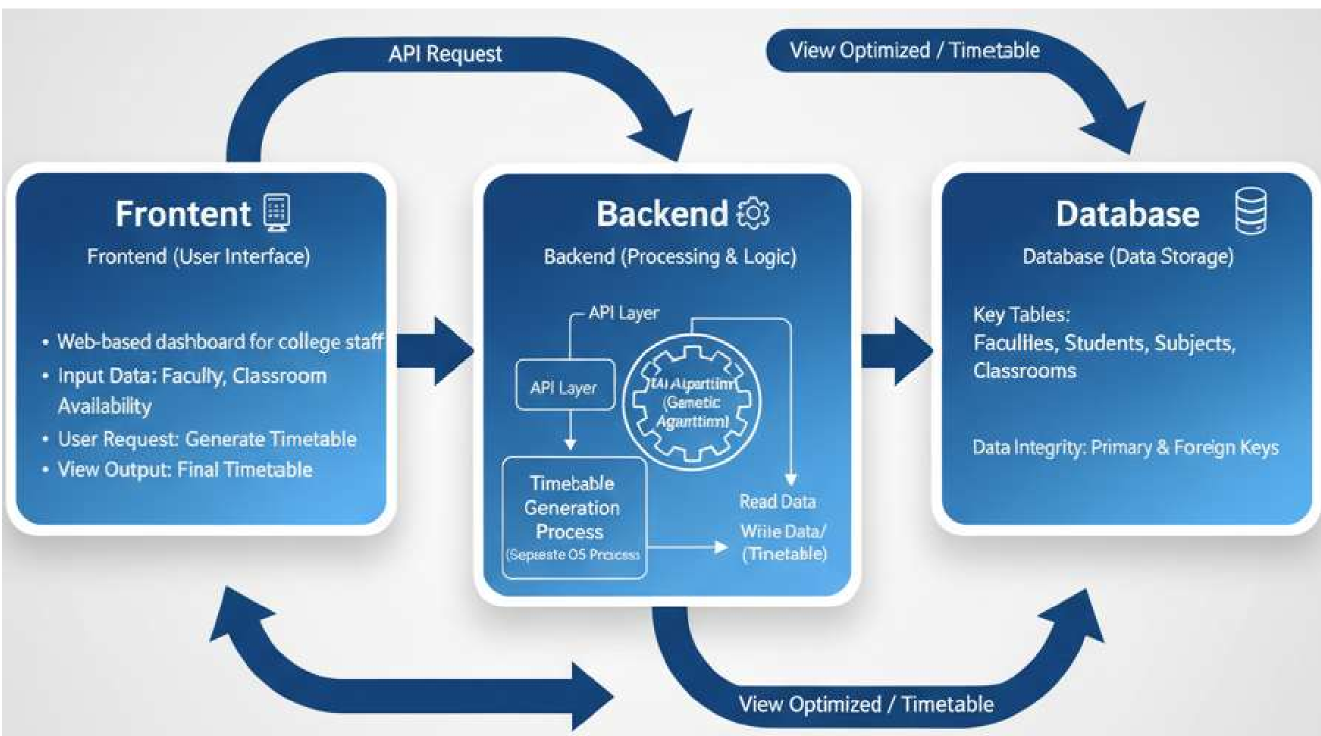
- **Phase 3: Finishing & Presentation (Month 3)**
  The final month is all about polishing the project and making it ready for a successful presentation.
  - **Milestone 3:** We'll focus on refining the user interface to make it easy and intuitive to use. We'll also conduct thorough testing to fix any bugs and ensure the system works smoothly. Finally, we'll prepare all the necessary documentation for both our OS and DBMS subjects, clearly explaining how our project applies the concepts we've discussed. At the end of this phase, the project will be fully ready to be presented.

Project Approach

Our approach is to create a professional web-based platform with a clear, three-part system. This design will help us build a strong and reliable solution that is also easy to manage.

1. **System Architecture and Design:** We will follow a well-defined approach with a focus on building a **scalable system**. The platform will have a frontend for the user interface, a backend for all the processing, and a database for storing information. The most important part of our design is the **AI-powered algorithm**, which will serve as the system's brain, making sure all timetables are as optimized as possible.
2. **Core Technologies:** For our backend logic, we will use **Python** with the **Django** framework, which is a secure and powerful choice for web applications. The core AI algorithm will be developed using a specialized library for **Genetic Algorithms**. To store all our project's data, we will use **MySQL**, a reliable relational database. For the user interface, we will build a modern and easy-to-use dashboard using **React.js**.
3. **OS and DBMS Integration:** This project is a perfect way to show our understanding of both **DBMS** and **OS** concepts. Our database design, which includes different tables for data and clear rules to keep it correct, will show our knowledge of DBMS. On the OS side, the timetable generation will run as a separate **process** in the background. This will keep the main website fast and responsive, demonstrating our ability to manage system resources effectively, just like a real operating system.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

When we're done with this project, we'll have a complete and smart system that's ready to solve real-world problems. We're focused on delivering two main things: the actual product and the paperwork that goes with it.

1. **The Final Product:** We're going to build a website that works great. It'll have a clean dashboard where college staff can easily put in all the info they need. The coolest part is the **AI-powered brain** we're building. It'll automatically crank out a perfect timetable with zero clashes. Basically, we'll be handing over a ready-to-use tool that does all the hard work.
2. **The Supporting Documents:** We'll also finish with a bunch of documents that show off our work. This includes a full **database map** (we call it a schema) that shows how we organized all the information. Plus, we'll write up a report that explains how our project uses all the stuff we've learned in our **OS and DBMS** classes. It's how we prove this isn't just a cool project, but a smart one too.

Assumptions

For this project to work, we're making a few simple assumptions. First, we assume the college will provide us with all the correct information we need, like all the classes, the number of students, and any special requests from teachers. Second, our project is designed for just one college or campus, not for multiple locations. Finally, we're assuming that our AI brain can actually figure out the best timetable for us. These simple ideas help us focus on our main goal without getting stuck on things that are outside our control.

References

For this project proposal, we looked at a bunch of resources and got some great help.

- **Dr. Noor Mohd.:** We talked to our college's timetable in-charge, Dr. Noor Mohd., directly. He gave us a lot of important info about the problems with our current timetable, which really helped us figure out what we need to solve.
- **Genetic Algorithms:** We studied how Genetic Algorithms work to understand how they could help us make a better timetable.
    - https://www.mathworks.com/help/gads/what-is-the-genetic-algorithm.html
    - http://deap.gel.ulaval.ca/doc/default/
- **Web Frameworks and Databases:** We looked at the official documentation for the tools we plan to use to make sure they were the right fit.
    - **Django:** https://docs.djangoproject.com/
    - **MySQL:** https://www.mysql.com/